COMP1801 Coursework exemplars

> Note that RED text like this represents words from the marker, while black text is from the exemplars written by former students.

This document will provide you with some examples of answers given for a **regression** question from a previous year's coursework. Similar examples for the classification question will not be provided (and it should be noted that in this old coursework there was a separate Neural Network question, so none of the solutions here try applying a NN to the data, only traditional methods).

Please note that these are responses used for a completely different dataset and that the current coursework's questions have been slightly rewritten, so what methods worked for them may not work for you and these solutions may not be written and/or presented in such a way that they perfectly address the task set for you this year. **The purpose of these exemplars is to give you a general idea of what is generally looked for in a write up of a ML solution – but you should not just uncritically copy their approach without thinking if something they have done applies to your problem!**

It should be noted that while all of these were considered passable solutions, none of them were perfect and plenty could have been done to improve all of them – so please review the marker's general feedback comments for each of the solutions.

Hopefully, these exemplars prove useful in helping you make a start on your own coursework.

Exemplar 1:

The given dataset was fitted with four different regression models, and these models were compared and assessed using a regression evaluation measure to ascertain which model performs the best in terms of the metric and more accurately forecasts a customer's compensation. The dataset is first imported, and data pre- processing which requires a number of steps is then carried out. Numerous columns in the dataset contain category and string data. The LabelEncoder() function is used to apply label encoding, which changes the values of the columns "Education," "WorkType," "Sex," and " Region " from category or string to numerical ones. In accordance with the alphabetically of the data for that specific column, an integer value is given to each category value. Because there are multiple categories for certain columns, like "Education," and because the order is irrelevant for categorical features, label encoder was chosen over one encoder. In contrast, employing a single hot encoder would have divided that column into several columns with just 1s and 0s as values, which would have resulted in excessive memory use. The dataset is then divided into a test set and train set. To prevent overfitting in the areas where the model is trained, splitting the dataset is an important step. Age, SiteSpending, SiteTime, RecommendImpression, Education, WorkType, Sex, Region, and RecommendImpression in this instance are the independent characteristics, while y is the dependent variable (Salary). As a result, y_train serves as the ground truth while X_train serves as the features for training the model. Similarly, x_test is utilised as a feature and y_test is used to confirm the anticipated labels during testing. Implementing polynomial features, which aid in recognising the curvilinear relationship between the dependent and independent variables, is the final stage in the data preparation process. In order to improve the model's accuracy, the means polynomial aids in identifying the closest-approximation link between both the independent and dependent variables. The degree of order 2 was selected since it was thought to be the best value for the fitting of the data. By creating a brand-new matrix column (polynomial combinations) for every one of them, where the values were produced by squaring X_train and x_test, the PolynomialFeatures() function was employed to fit and convert the X_train and x_test. X_train_polynom and x_test_polynom now contain the new set of data, respectively. Once data preparation is complete, the simple linear regression model is presented using the LinearRegression() function, where the fit intercept parameter is set to true, indicating that the line of best fit will decide the y-intercept. After that, the model is trained with the help of the variables X_train poly and y_train, and to generate predictions with the help of the testing set, x_test poly is employed by feeding it into the function predict() and storing it in variable y_pred. The y values of the testing set x_test_polynom and predicted y values from y_pred were then shown on a scatter plot of age against pay. The output gotten is R-squared and Root Mean Squared error to assess the model and mean squared error() methods, the R-squared and Root Mean Squared error were calculated.

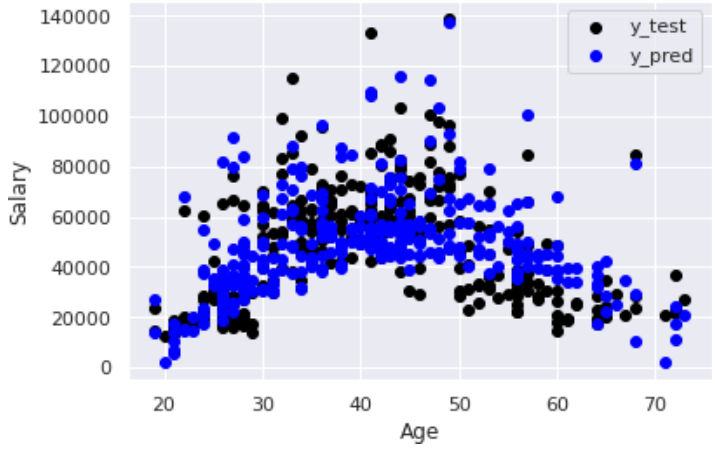


Figure 1: Scatter Plot Salary vs Age.

As liner regression I did 3 more different regressions (Lassos Regression, Ridge Regression, Random Forest Regression). The outputs I got from all 4 regressions are provide in below table (table1:).

Model	R-squared score	RMSE
Simple Linear Regression	0.6988	162074221.3697
Lasso Regression	0.6988	162073978.8304
Ridge Regression	0.6988	162073239.0961

Random Forest Regression	0.8653	72506294.1483

Table 1: Regression Model outputs.

The standard error of the regression decreases as adjusted R-squared increases when comparing regression models with the same dependent variable and estimate period. As a result, one may use adjusted R-squared as a ranking criterion since the model with the greatest adjusted R-squared will also have the lowest regression standard error. In our case, the Random Forest Regression model has the lowest regression standard error since it obtained the highest R-square value based on table no 1. Where simple liner regression, lasso regression and ridge regression values are quite of similar. As, the accuracy level is high in Random Forest Regression data, it will be the best fit model to predict the customer's salary.

Comments to improve:

- Text can be a little hard to follow because there are no paragraphs. Remember you do get marks for presentation and English quality.
- While it is good that multiple models were tried, there fact that Linear, Lasso and Ridge are all the same should have rung alarm bells – clearly the regularization term is too strong.
- There is no information on the hyper-parameter used either for regularisation or any of the random forest hyperparameters. No discussion of how these values were chosen or if they were even tested.
- Not explained how or why the polynomial degree of 2 was picked.
- Do not clearly explain why these metrics (R2 and RMSE) were chosen and what they mean.

Exemplar 2

Before it was possible to work with the data that was given it was needed to load the data into the data frame from the uploaded .csv file. It was possible with files.upload() that asks us to upload the CSV file and then the CSV file is saved to a pandas data frame. After that, it was necessary to make sure that all of the data can be used in a machine-learning algorithm. By using the LabelEncoder it was possible to encode the object data types in the given .csv file into the float and that way all data can be used in the Machine learning models to predict the salary. The Results for this task were measured by using the R2 score also known as the coefficient of determination, which is a measure of the goodness of fit of a regression model. It is calculated as the ratio of the variation in the dependent variable which is explained by the regression model, to the total variation in the dependent variable. The R2 score can range from 0 to 1, where a higher score indicates a better fit of the model to the data. The data was split using the train test split function with a test size of 0.25 and a random state of 0. After seeing that the run with just Linear Regression without finding the best degree and using poly results were 30% of accuracy it was decided to use functions to find the best degree and best alpha. For this task, the three traditional machine learning models were chosen. The first one is the Linear regression model. Linear regression model a machine learning algorithm used for classification tasks. It is called logistic regression because it uses a logistic function to model the relationship between the dependent variable and one or more independent variables. This allows the model to predict the probability that a given input belongs to a certain class, based on the input features. The R2 score that was acquired was 0.6239. Meaning that the model could predict 62% of the salary. This is a significant increase in the accuracy compared to the basic linear regression fitting with the increase being from 30% to 62% of accuracy. Therefore the next models for the experiment were already done with the best alpha calculations to increase the accuracy of the predictions. The second model that was used to compare to the first one was the Ridge model. It is another type of regression analysis that uses a regulation term to penalize the magnitude of the regression coefficients. The penalty term is called the L2 penalty, which helps prevent overfitting and improve the interpretability of the model by reducing the complexity coefficients. The R2 results using the Ridge model was 0.5576 meaning that the Ridge model could predict only 55% of the salaries. The third model that was used to compare to the first two was the Lasso model if compared to Ridge, Lasso uses L1 as the penalty term that helps to prevent overfitting and to improve the interpretability of the model. The R2 results that were obtained by fitting the Lasso model were 0.7186. Meaning that 72% of the salary was predicted correctly. The clear leader that is chosen for the prediction of the salary is the Lasso regression model. As it could achieve 72% of the salary predictions right. Therefore the Lasso model is chosen to perform the predictions for the salary on this given dataset.

Comments to improve:

- This is quite short, but it does include all of the key aspects – such as evidence of experimentation to improve the models, clear reporting of the results and a description of the evaluation metrics used.
- Also hard to read as a single block with no paragraphs.
- No images or tables are included, which could have presented many aspects of the results in an easier to digest way.
- There is discussion of using a function to find the best regularization strength, however:
    - Results of this experimentation not reported (e.g. in a table showing the accuracy for different values of alpha)
    - The best regularization strength (or any other hyperparameters) were never recorded.
- There is an error where the report talks about logistic regression for a sentence, assumingly some sort of typo/copy paste error.

Exemplar 3

Regression models are a type of models fitted when there is a numerical target wanted to be predicted, in this case, Salary. The type of regression to be fitted will be multivariable as there are several different features.

We will be using Mean Squared Error (MSE) as a cost function to compare performance in models in validation data using steepest descent (SD) algorithm.

$$MSE = L(\theta) = \frac{1}{m}\sum_{i=0}^{m-1}(y^i - h_\theta(x^i))^2$$

The goal is to minimize MSE on validation data, for that, SD will be performed with different regularization weights (α). These alphas are very important as choosing the wrong one can led to over fitting or under fitting.

Complex models for regression can be applied, like polynomial regression. However, this usually leads to overfitting, especially when there are multiple features as the model tends to be too complex. These complex models always need regularisation to avoid overfitting.

$$J(\theta) = loss\ function + \lambda * regularisation\ term = L(\theta) + \lambda R(\theta)$$

This metric (MSE) is useful to compare models for different regularization strengths (α), however, it is not an absolute value that could be used to determine how well the model is performing, compared with other models with different algorithms implemented, as it depends on the scale of the features. For this reason, the evaluation metric of choice is r-squared (R^2), a number from 0 to 1 that tells the percentage of variation that the model can explain. R^2 of 1 would be a perfect relationship and 0 would be randomness.

The model with a higher R^2 is a multiple regression using all features (categorical and numerical) and polynomial features too with a regularization model of Lasso. (see GoogleColab 1.3.2). For this model, Label encoding was used for the variables Education, WorkType, Sex and Region. After that, polynomial features were created. The maximum degree of choice was 5, higher degree will quickly scale to a huge number of variables in the model. MSE for different regularization weights was stored in both validation and training data set, concluding the best alpha to be 100, as we can see in the graph below.

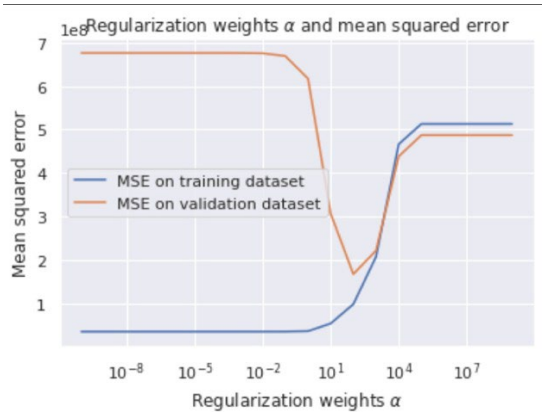


Figure 1: MSE on different Regularization weights on training and validation data.

To have an idea of how the model did, a graph of the target against one of the features used in the model is displayed. Please note the graph below represents only one feature, in this case Age, however the overall performance of the model is measured using many more features that might or might not have performed as well. Find in black the true salary in the test data set and in blue the model prediction for each age.

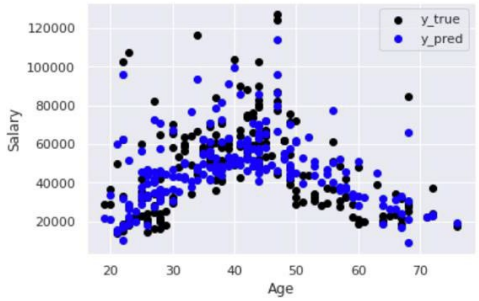


Figure 2: Salary against Age. Compare the true target value with the prediction by the model.

The R^2 for this model is 0.7385. This means the model was able to explain almost 74% of the variation of the data, which is an acceptable score, and the model could be used for predictions with quite high accuracy. The same model using Ridge regularization instead of Lasso achieved $R^2 = 0.6287$. (see GoogleColab 1.3.1)

Comments to improve:

- Nice to have the background detail on the MSE and regularization, not strictly required, but I feel it supports the point of going straight in with a regularized regression.
- Would have been nice to have a bit more detail on the ridge implementation (e.g. the learning curve)
- Should refer to figures by name (‘Figure 1’), rather than referring to as “the graph below”
- Some data is missing, such as the data splitting used and what type (if any) feature scaling was used.
- Not made totally clear that all features were used.
- While going straight to regularized regression is justified, I think it would have been nice to see the results for unregularized regression, to demonstrate more experimentation and give a nice selection of results to put in a table.
- Could have tried one hot encoding

Exemplar 4

Regression in Machine Learning works by predicting a real value of target by learning from the features and fitting best fit line for the data which minimizes the overall cost function. Regression can be linear based on only one feature, or it can have multiple features to make predictions. In this case, the prediction value is yearly salary of customer in pounds (GBP).

The general approach to use regression for prediction is to fit the best line with minimum squared distance between the datapoints avoiding overfitting and to use this fit to make prediction of new data points.

In this case, the dataset is already cleaned so no major pre-processing is required. Label encoding is performed on the categorical features which are – Education, WorkType, Sex, and Region which coverts category string values into integer values which can be used for regression. Next step is to split the data into train test and validation split using 80 percent data for training process. Data is shuffled randomly for better results as it is not time series data and dropping index as it does not have any effect on the data.

The next step is to fit a basic multivariate regression model without tuning any hyper parameters and using all the features to get the intuition on how the model performs on just the data with 80 percent training data and analysing further steps to take. The model does not perform well and gives very high mean squared error and a R2 score of '0.25', which on plotting gives results as observed in the *figure 1*, showing black points as true value and blue points a s prediction.

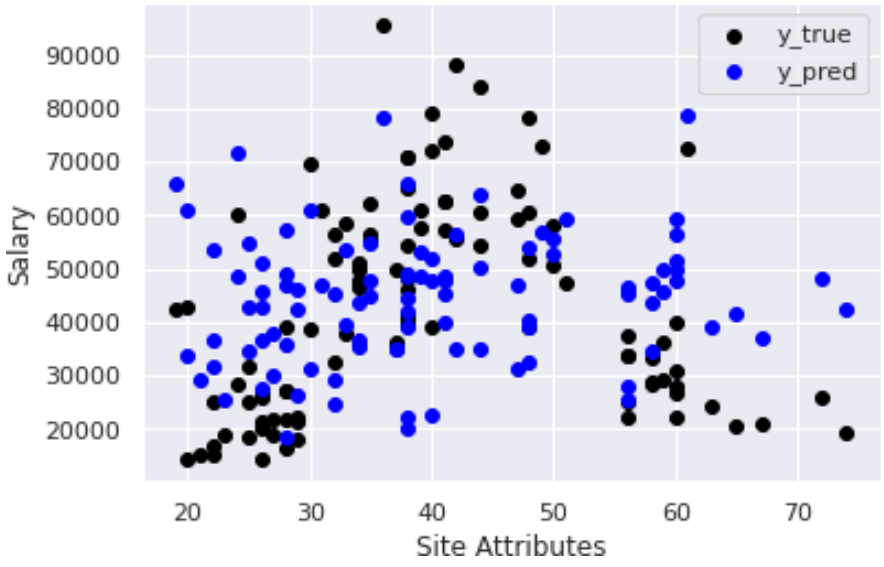


Figure 1 - Linear Regression Salary Prediction

Since the model does not perform well, the next step is to experiment with different regression model accounting for feature scaling, polynomial features and to use regularization penalty to get the best model with least mean squared error and high value of R2 score.

Using polynomial features and Standard scaler on features is giving better performance in comparison due to which both are applied to the training data. The experimentation to get the best regression model is based on the mathematical intuition of using penalty factor to account for features which does not contribute much for prediction and the way it is done is by using L1 and L2 regularizations and finding appropriate value of Learning rate (Alpha) for the model. This is done by repeatedly testing different values of learning rate on lasso regression model in such a way that the best value is obtained in the end.

Therefore, on finding best value for degree, using lasso regression, and looping through different values of Learning rate to test and find best alpha value, the performance of model improved significantly which gives

R2 (R squared score) value of '0.7452' which means that the performance of this model is best compared to previous iterations and *figure 2* gives the plot of predictions.

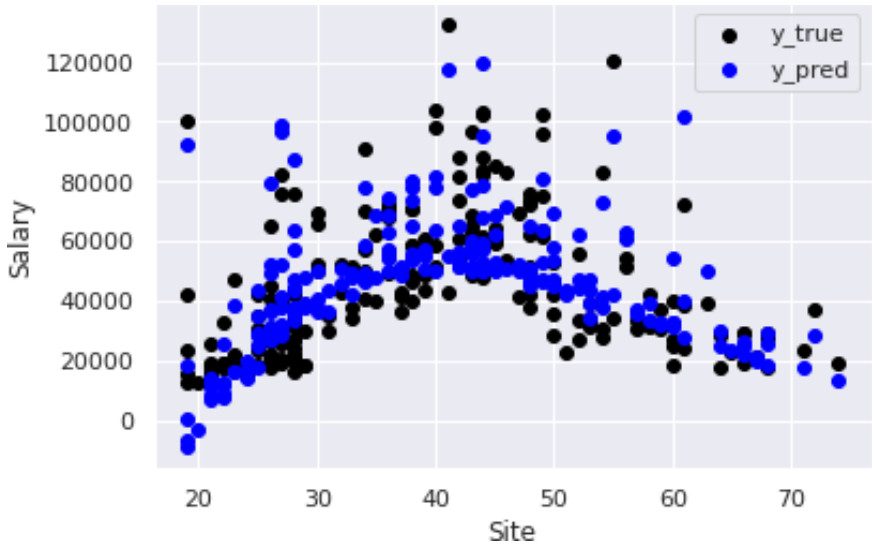


Figure 2 - Lasso Regression Predictions

The best model uses Lasso Regression with best learning rate of 8.0, polynomial degree of 3, and uses only 4 features which are – Age, Site spending, Site time, and Recommend Impression as appropriate features with standard scaling and random state and gives results as observed in *figure 2*.

This model gives the least value for mean squared error and highest value of R2 score which is '0.7452' in comparison to simple linear regression model, ridge regression model and other models with using less or different features due to which this model on experimentation due to which it is termed as the best model for prediction.

R2 score or coefficient of determination is used as the regression evaluation metric to get this model as R2 score gives the best idea regarding the variance in predictions. In simple terms, R2 score measures how much error in prediction is eliminated, by calculating the reduction in error in percentage terms. A high percentage value of R2 indicates that prediction error is minimized. In this case, score is '0.7452' or nearly 75 percentage which tells that model performance is very good in comparison to previous results with less R2 score. Following Table (Table 1) summarizes the R2 score performance of different models tested for *salary* prediction.

Model	**R2 Score**
Linear Regression	0.2577
Ridge Regression	0.7432
Lasso Regression	0.6909
Final Lasso Regression	0.7452

Table 1

Comments to improve:

- While full hyperparameters are given for the final model (that is the most important thing), it would be good to know the other hyperparameters used in the other models in table 1.
- No justification/proof of testing for why we are only using some of the features.
- Could provide more detail of how the best hyperparameters were chosen (training curves, etc).
- Not clear what type of categorical coding was used or if this was experimented with.
- Data only appears to be split into training and test sets – no validation set. Unclear what the procedure was for model selecting then, as the test set should have been retained for evaluating the FINAL model. This omission does throw through the accuracy of all the results into question.

Exemplar 5

The traditional machine learning algorithms namely, Lasso, Ridge, Linear Regression, and Random Forest regressor were fit to the data. The regression algorithm chosen was Random Forest regressor. The implementation for the Random Forest regressor is as follows:

Data pre-processing:

Initially, the data consisted of categorical values; these values were converted to numerical values using label encoding. All features were used to fit the model. The features included were age, site spending, site time, recommend impression, work type, sex, region, and education.

Data splitting and Feature scaling:

The data was split using an 80:10:10 ratio. 80% of the data for training, 10% for validation, and 10% for testing. The use of this split gave the model more data to train on, and hence it performed better as compared to using 60:20:20 or 70:15:15 splits. As each feature in the dataset had a different scale, standardization was performed using standard scaler to get the features to have a common scale.

Model training and Hyperparameter tuning:

To uncover any non-linear relationship between the data, the features were scaled to a polynomial degree in the range from 1 to 5.

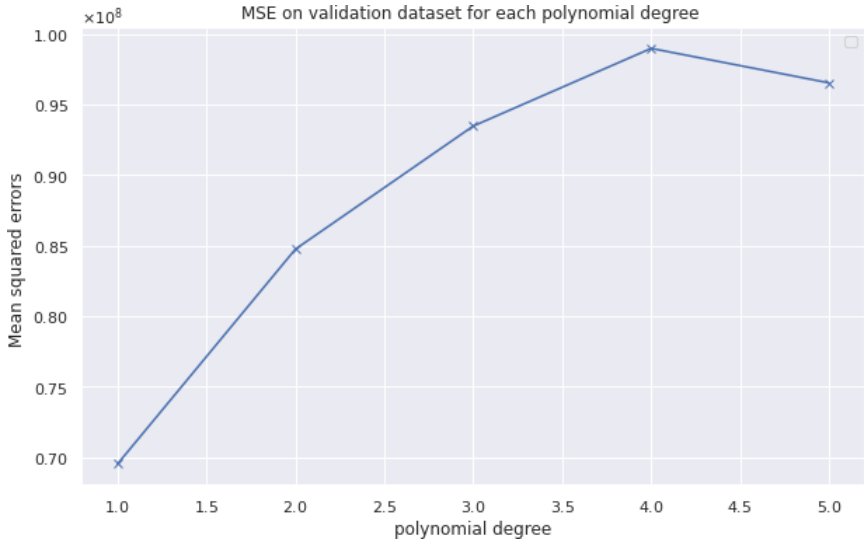


Figure 1. MSE on validation dataset for each polynomial degree

The graph in Figure 1 indicates the Mean Squared Error (MSE) on the validation data set for each polynomial degree. As seen in Figure 1, polynomial degree 1 has the lowest MSE, which is 0.69528634 x10^8on the validation dataset and is chosen as the best degree for the model.

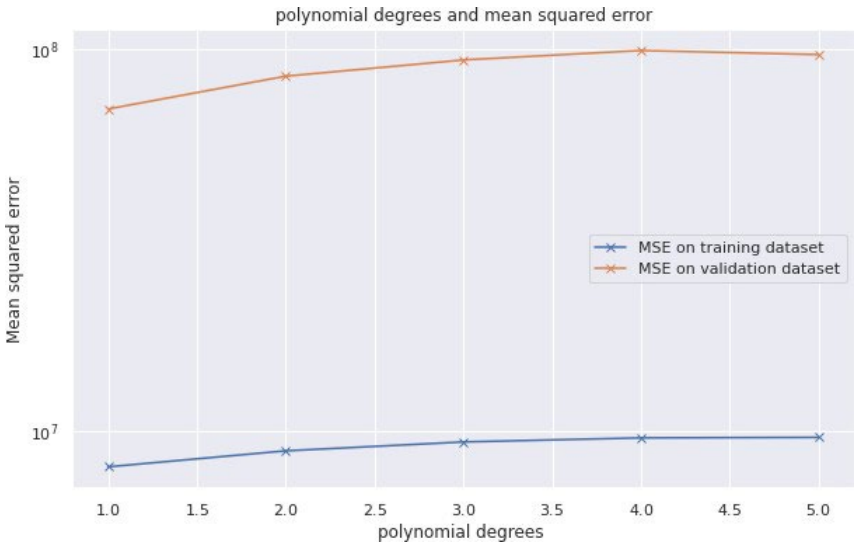


Figure 2 Training and validation MSE for each polynomial degree

Figure 2 depicts the relationship between MSE on the training and validation datasets for each polynomial degree. As seen in the graph above, there is a much higher MSE on the validation dataset, indicating possible overfitting on the training data set.

Model testing and comparison:

The model was fit again using the training dataset with the best selected polynomial degree of 1, n_estimators of 200, and the default criterion 'squared_error' was used. The model was made to predict values on the test dataset, achieving a MSE of 0.64727185 x10^8 and an R squared value of 0.8889. The MSE generated on the test data (0.64727185 x10^8) is much lower than the MSE generated on the validation data (0.69528634 x10^8), which indicates a good performing model. Figure 3 shows the predicted values versus the true values between age and salary of customers. In Figure 3, the model's predicted values are not far from the true values in most cases, indicating a well performing model.

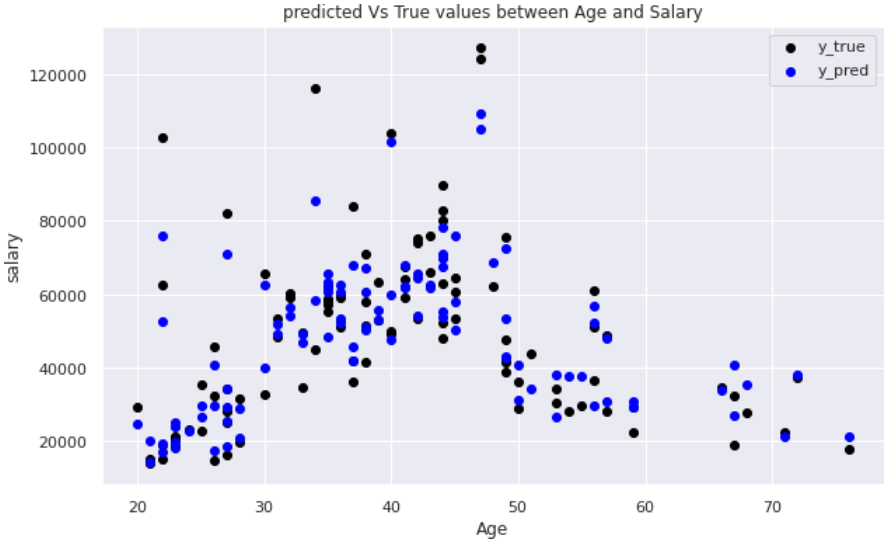


Figure 3. Predicted v/s True values between Age and Salary

Table I. Regression model comparison based on MSE on test dataset and R squared score

Models	Hyperparameters	Mean squared error (MSE) on test dataset	R squared score on test dataset
Random Forest Regressor	degree=1	64727185.46176836	0.8889
Lasso Regression	degree=2, regularization weight=100	109027366.80854566	0.8129
Linear regression	degree=3	134632869.23118144	0.7689
Ridge regression	degree=2, regularization weight=1	146368866.98240042	0.7488

In Table I, the mean squared errors were recorded on the test dataset after training each model with the best possible polynomial degree and regularisation weights in the case of Lasso and Ridge. As the models are fitting the same data and predicting the same target, the MSE can be compared between them. From the table above, the MSE of the Random Forest regressor is the lowest, while it also has the highest R squared value. Using these parameters for evaluation, the model selected is the Random Forest regressor model for performing regression.

The percentage of the variance in the data points that can be explained by the regression line or model is known as R-squared, R2, or coefficients of determination (Kumar, 2022). The value of R-squared can range from 0 to 100%. The value of R-squared as seen in Table I for the Random Forest regressor is 0.8889 meaning that the model accounts for 88% of the variation in the data. Since the value is greater than 0.70, one can conclude the model is a good performer.

Comments to improve:

- The section looking at polynomial degree seems a bit irrelevant, as we end up testing that and using other degrees in our model anyway.
- There are other hyperparameters for a random forest model that have not been discussed.
- Did not try one hot encoding.
- Could have used more detail about how the experimentation was undertaken – was a grid search used? Manual? A self coded loop?
    - We are simply given the random forest regressor with no detail of why we chose this model and how its parameters were chosen.
    - Unclear if the other models in the table are the 'best' models the student were able to obtain after experimenting for each mode. Again, the process for obtaining these models was not made clear.
    - Ideally we want a clear explanation of what experimentation was done, why it was done (including why it was done in any particular order) and there should be some proof provided as to the results.